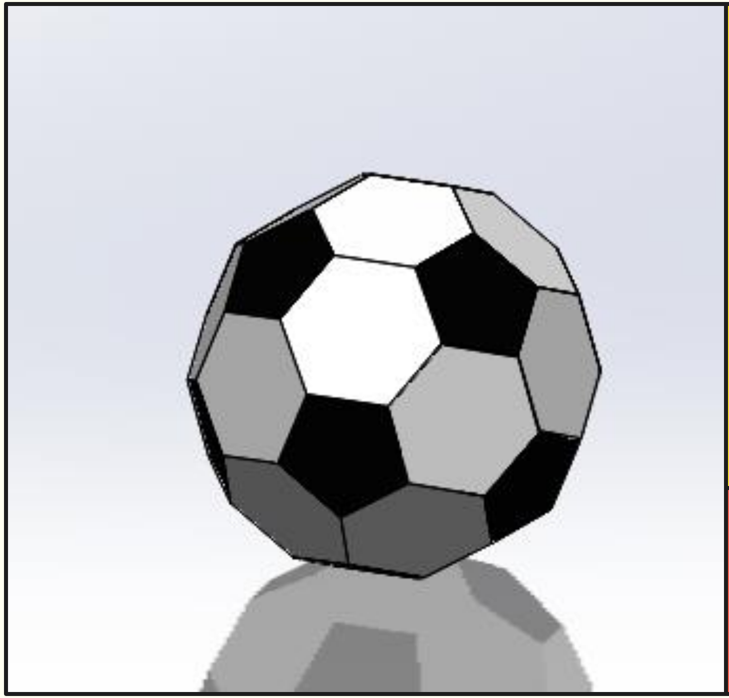
10/24/25
Friday
MAE 1001

Solidworks Crash Course 1!

Your Gateway to 3D Creativity!

Slides By Nathan Janssen 2025 for MAE 1001

Session Outcomes



Create 1/70th of a Soccer Ball!

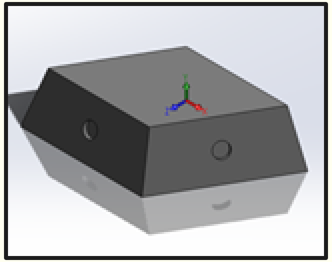
Today, we're going to introduce:

- **2D Sketching** a Part
- Converting 2D Sketches to a **3D Part**
- The Solidworks **Drawing feature**

These three skills will help you start your **professional career.**

At the end of next week's session, you will have a **customized** piece of this soccer ball!

Your Part!



Classrooms To Access Solidworks:
Remotely and Phyiscally

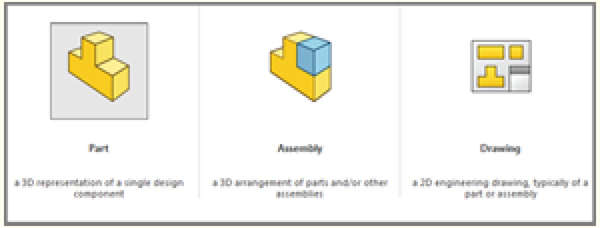
SEH 3040	SEH 4040	TOMP 405	TOMP 406

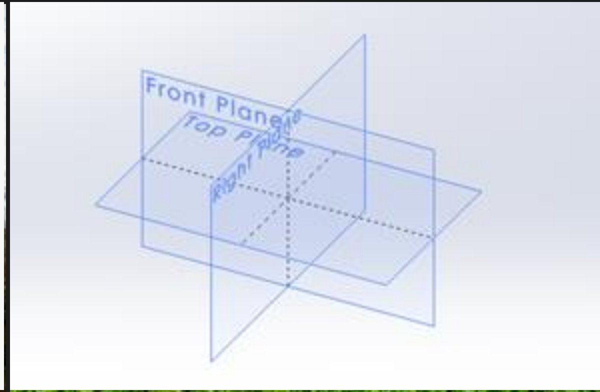
Classes where you'll see Solidworks

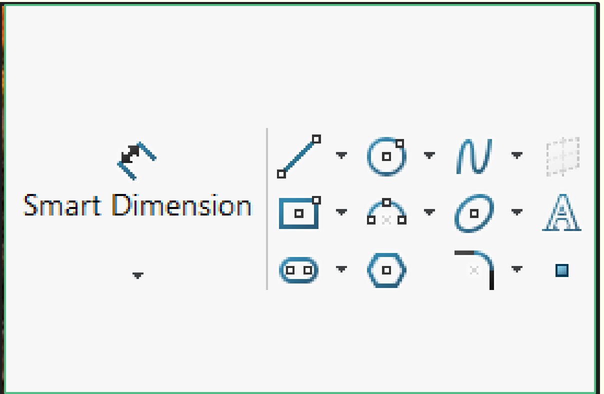
MAE 1001	Introduction to Mechanical and Aerospace Engineering
MAE 1004	Engineering Drawing and Computer Graphics
MAE 3187	Heat Transfer
MAE 3191	Mechanical Design of Machine Elements
MAE 3192	Manufacturing Processes and Systems
MAE 3193	Mechanical Systems Design
MAE 4151	Capstone

And many more…
(as well as clubs, research, internships, etc.)

Part 1: 2D Sketches







1. Start a Sketch

This is the first step when modelling any part. It creates a 2D drawing that, using the wide variety of tools given by solidworks, can be made into any complex 3D shape.

2. Select a Plane

Orienting yourself in 3D space is crucial in solidworks, especially as you work with more and more complex models and assemblies, make sure that you know what plane you're working on.

3. Start with the basics!

Solidworks offers a variety of 2D modelling tools to help you get starting in your modelling journey. Play with these and get familiar with them as they will allow you to make more complex parts.

Stuck? Hint, check your units! 

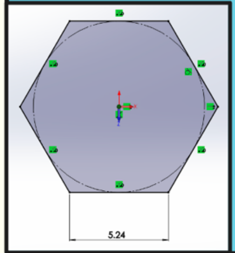
Let's Make our Part!

<u>Hint</u>: **Follow** along here and feel free **ask questions**, when we're done with this part you'll be **75% done** with your **CAD assignment**!

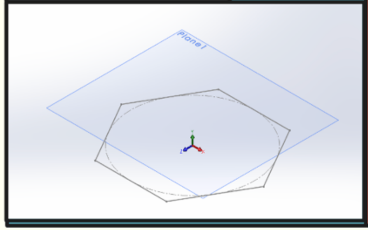
Janssen 2025

1.

First, let's create a hexagon using the (polygon) tool in the sketch tab.
Then, a reference plane offset by 2cm.

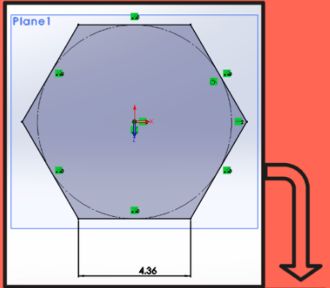


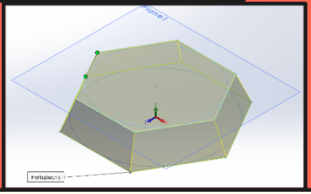
Reference Plane Tool!



2.

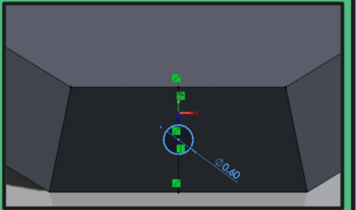
Next, on our reference plane, make another, smaller hexagon.
Use the Lofted Boss/Base tool to connect the sketches.

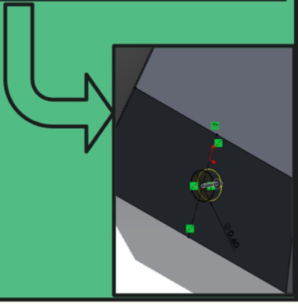




3.

Now, centered on one face of our truncated hexa shape, sketch a circle and Extruded Cut the 2cm into the part

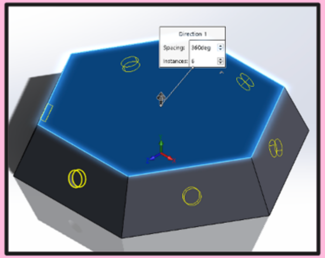




4.

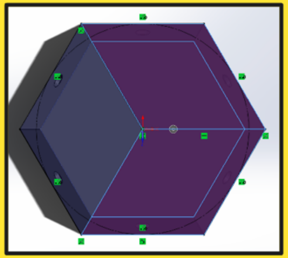
Home stretch!

Use the Linear Pattern drop down menu and find Circular Pattern and select the top face.



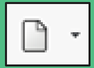
5.

We did it! But we only need ⅓!
On the top face, sketch three evenly sized pieces and don't forget to redraw the boundary. Then Extrude Cut through everything that you don't need!



Making our Part into a Drawing!

First,

From the part, find the  (New) dropdown and select the Drawing option

Then,

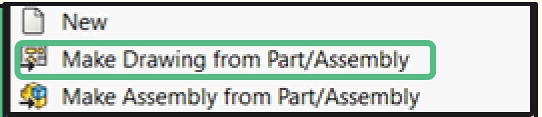
Turn off the standard formats option and find ANSI B (a common format)

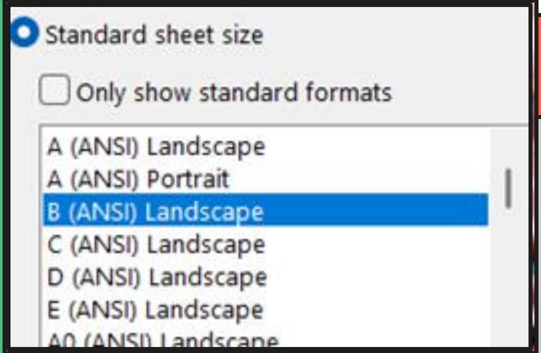
Next,

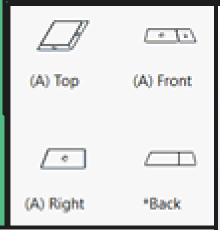
Select the front face of the part and drag it to the bottom left of the drawing

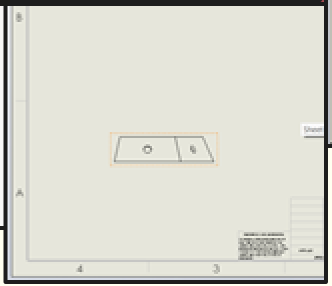
Finally,

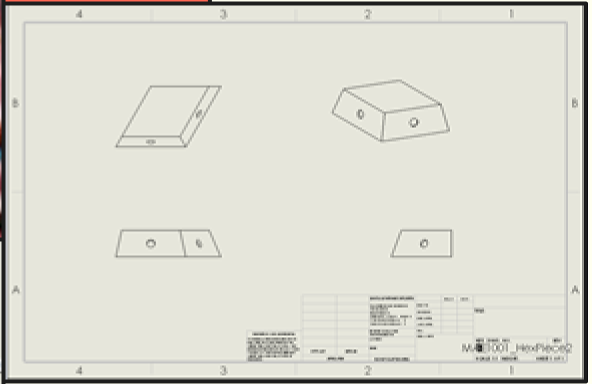
Solidworks *automagically* populates where you move the part to the side that should be viewed, click around the front face to finish your drawing.











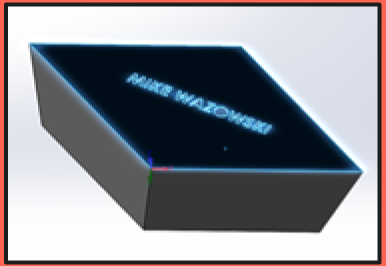
Customization Criteria

No Extrusions over 2cm!

Let's try to conserve our materials the best we can and not be wasteful while still having fun!

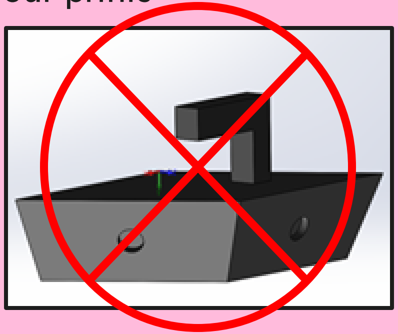
Your name must be on it!

Use the (Text) tool to add your name to the drawing and either cut or extrude it out ~0.2 cm



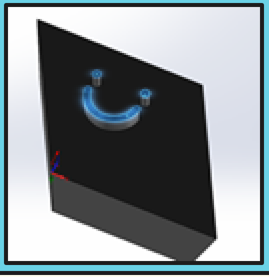
No Overhangs!

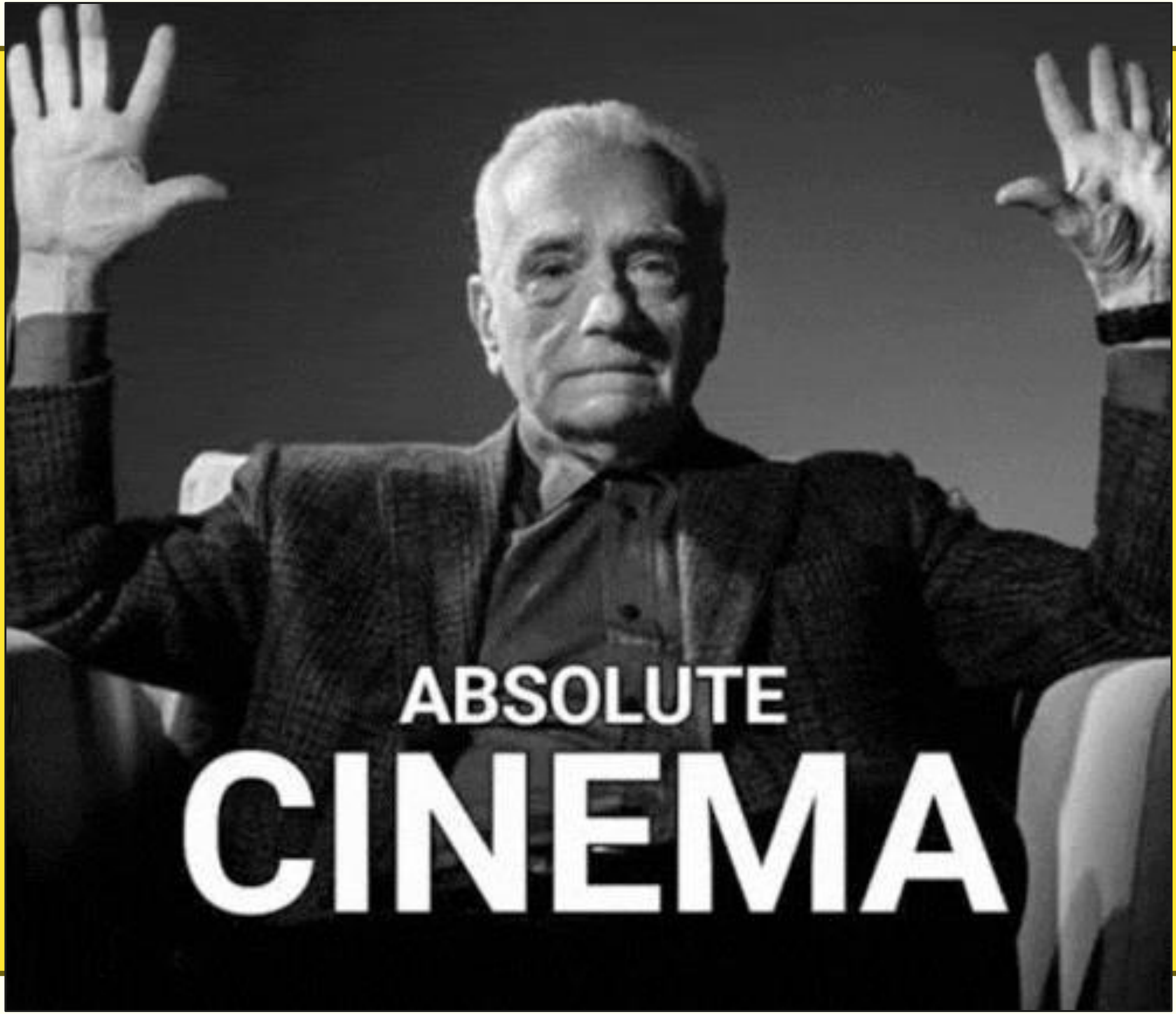
We are trying to print with **NO SUPPORTS**, we can do this as long as we have no overhangs on our prints



Have fun and be original!

Remember, this is supposed to be fun! Try something new, if it doesn't work, ask me, or a friend, or the internet.




ABSOLUTE
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Q&A

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